

PROJECT REPORT

SPOTIFY STREAMING ANALYTICS DASHBOARD

Data Visualization using Microsoft Power BI



Submitted by

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Abstract

This project presents a Spotify Streaming Analytics Dashboard developed in Microsoft Power BI. The dashboard is designed to convert streaming-related data into a visual and interactive report. It brings together summary indicators, track-level comparisons, monthly activity, weekday usage, and filtering controls in one place. The visual theme uses Spotify-inspired green, black, and white to keep the interface consistent with the music platform.

The report explains the project objectives, dashboard components, data preparation workflow, visualization choices, key observations, limitations, and possible future improvements. The screenshots provided for this report are used as the design reference. Values shown in the dashboard should be validated against the original dataset before they are treated as official findings.

1. Introduction

Music streaming platforms generate data about tracks, artists, release dates, streams, and audio characteristics. Reviewing such data in raw tables can make it difficult to identify popular tracks, changes over time, and differences between songs. Data visualization helps present these details in a clear format so that users can explore the information and compare categories quickly.

The Spotify Streaming Analytics Dashboard uses Power BI visuals to provide a single-page overview of selected streaming metrics. It is intended as an academic data-visualization project rather than an official Spotify product or a verified statement about Spotify-wide listening behaviour.

2. Problem Statement

Streaming data may contain many records and several types of measures. Without a structured visual report, users may find it difficult to compare track performance, review monthly patterns, and filter records by artist, track, release date, or year. The project addresses this challenge by organizing the data into an interactive dashboard with summary KPIs and complementary charts.

3. Objectives

- Present important streaming metrics in a compact and readable dashboard.
- Compare the stream counts of selected tracks.
- Show month-wise track counts and weekday-based usage patterns.
- Provide slicers for Artist, Track, and Release Date, plus year-selection controls.
- Use consistent colors, labels, and chart types to improve readability.
- Practice data preparation, visual design, and reporting using Microsoft Power BI.

4. Tools and Technologies

Tool / Component	Purpose
Microsoft Power BI	Data modelling, measures, slicers, and dashboard visuals.
Dataset (source file to be documented)	Provides track, artist, date, stream, and audio-feature fields.
Power Query	Can be used to clean columns, correct data types, and handle missing values.

5. Dashboard Design and Components

The dashboard is arranged into a top filter/navigation strip, summary and featured-track areas, a central detail section, and three lower charts. The dark background with green highlights provides visual consistency. White labels are used for contrast against the dark panels.



Figure 1. Full Spotify Streaming Analytics Dashboard (provided screenshot).

5.1 Filters and navigation

- Artist slicer: narrows the report to the selected artist.
- Track slicer: focuses the report on a selected track.
- Release Date slicer: filters records by the available release-date values.
- Year buttons: allow year-based selection for the displayed period, if configured as interactive slicers.
- Reset/back-style icon: intended to return the report to a default view; its actual behaviour depends on the Power BI action configured.

5.2 Summary cards and featured track

The top section displays an average-stream measure and a comparison labelled “Top VS Avg,” alongside a featured-track panel. The green welcome panel presents selected track, artist, stream, artist-count, and audio-feature indicators. These cards are useful for quick scanning, but their titles, units, and aggregation logic should be clearly defined in the final Power BI file.

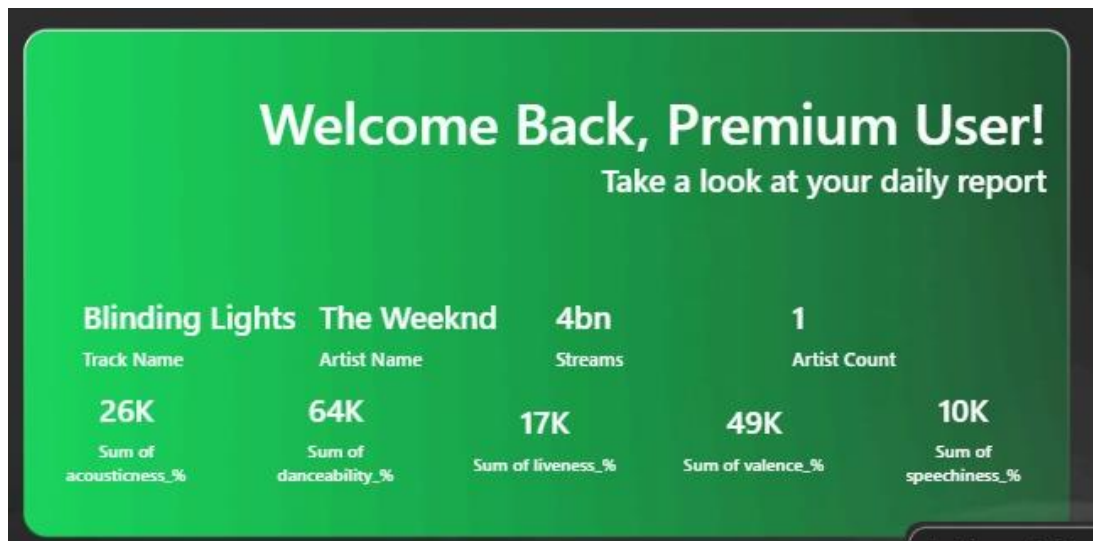


Figure 2. Featured-track and KPI panel.

5.3 Charts and their purpose

Visual	What it shows	Why it is useful
Count of Track Name by Date (line chart)	Month-wise change in the count of tracks.	Helps spot higher and lower months.
Spotify Track Usage (matrix/heatmap)	Usage intensity across weekdays and months.	Makes repeated patterns and gaps easier to notice.
Spotify Stream by Track (horizontal bar chart)	Stream totals for the displayed tracks.	Supports quick ranking and comparison.

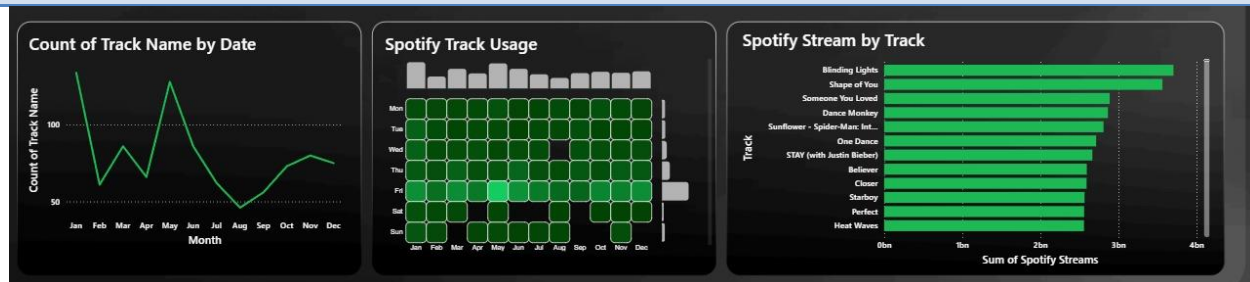


Figure 3. Monthly line chart, usage matrix, and track-stream comparison.

6. Data and Methodology

The dashboard design suggests that the underlying dataset includes fields such as track name, artist name, release date, stream count, and audio-feature percentages. The exact source, row count, date coverage, and field definitions were not supplied with the screenshots; therefore, this report does not claim a specific dataset origin or a verified sample size.

1. Import the source dataset into Power BI and record its source, download date, and licence/usage conditions.
2. Inspect column names, data types, missing values, duplicates, and inconsistent track or artist labels.
3. Convert release-date fields to a valid Date type and stream counts to a suitable numeric type.
4. Confirm whether stream totals are per track, per period, or repeated across multiple records before summing.
5. Create measures for the KPIs and test them against a few manually checked rows.
6. Build slicers and visuals, then verify that all visuals respond correctly to the selected filters.

7. Format titles, units, data labels, and number display consistently; document any exclusions or assumptions.

7. Key Observations from the Dashboard Layout

Based on the visible design, the bar chart places “Blinding Lights” near the top of the displayed track ranking, followed by other tracks such as “Shape of You” and “Someone You Loved.” The monthly line chart shows visible variation across months, while the usage matrix highlights differences across weekday/month combinations. These are observations from the screenshot only; exact rankings and conclusions must be confirmed from the dataset and the measure definitions.

- The horizontal bar chart is appropriate for comparing many track names because labels remain readable.
- The line chart emphasizes change across the months, but its title should specify the year or period.
- The matrix/heatmap provides a compact way to compare usage across two categorical dimensions.
- Slicers can make the dashboard more useful for exploring an artist, a track, or a release period.

8. Findings and Interpretation

The dashboard demonstrates how several related views can be combined to support exploration of streaming data. Track ranking gives a quick comparison of selected songs; the monthly view can help identify periods with higher or lower record counts; and the heatmap can reveal combinations that deserve further checking. However, track count and stream count are different measures and should not be interpreted as interchangeable.

The KPI panel includes values for audio characteristics such as acousticness, danceability, liveness, valence, and speechiness. Before presenting these as percentages, confirm whether the original fields are stored on a 0–1 scale, a 0–100 scale, or are being summed. For most audio-feature fields stored on a 0–1 scale, an average is usually more interpretable than a sum.

9. Limitations and Data-Quality Checks

- The original dataset and its source citation were not included with the screenshots.
- Some KPI labels and values may need clarification; for example, an audio-feature percentage should have a clearly defined aggregation and unit.
- The “Top VS Avg” percentage requires a documented formula and a clearly named comparison baseline.
- A year button or reset icon works only if the relevant Power BI interactions/bookmarks are configured correctly.
- A screenshot alone cannot confirm filter behaviour, data accuracy, refresh settings, or whether totals are duplicated.

Recommended validation: reconcile total streams with the source data, test slicers one by one, verify date boundaries, inspect duplicate rows, and ensure the same measure is used consistently across cards and charts.

10. Future Scope

- Add a documented data-source link and a short data dictionary.
- Include a date-range slicer and a clear “Reset Filters” button.
- Add tooltips showing track, artist, release date, and stream totals.
- Create drill-through pages for artist and track-level analysis.
- Use explicit units (K, M, B) and consistent decimal precision across all KPIs.
- Add data refresh details and a small methodology note inside the dashboard.
- Compare year-on-year streaming totals only when the dataset contains complete and comparable periods.

11. Conclusion

The Spotify Streaming Analytics Dashboard is a practical example of using Microsoft Power BI to present music-streaming information in a compact, interactive format. It combines KPI cards, slicers, a monthly line chart, a weekday/month usage matrix, and a track-ranking bar chart. The project supports learning in data preparation, measure design, visual selection, and dashboard layout. To make the final submission more reliable, the dataset source, KPI formulas, units, and filter interactions should be validated and documented.

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